Statistical Analysis of Student Performance in Mathematics and Language

Application Report

Author: Treva Antony Ogwang

Table of Contents

1. Introduction 3

2. Data and Research Questions 4

3. Statistical Methods 5

   3.1. Descriptive Statistics and Statistical Graphics 5

   3.2. Welch Two-Sample t-Test 6

   3.3. Cohen's d 6

   3.4. One-Way ANOVA 7

   3.5. Tukey HSD Post-Hoc Test 8

   3.6. Eta-Squared 8

   3.7. Assumption Checks 8

4. Descriptive Results 9

5. Gender Differences in Mathematics and Language 11

6. Parental Education Differences in Mathematics and Language. 12

7. Summary and Discussion 14

Bibliography 16

1. Introduction

Student performance is often influenced by several background factors. In educational data analysis, it is useful to examine whether differences in academic outcomes are related to variables such as gender or family educational background. Such analysis does not explain all reasons behind student achievement, but it can help identify patterns that may be relevant for further investigation.

This report analyzes the performance of 486 students in two subjects, Mathematics and Language. The main aim is to examine whether the scores differ by gender and by parental level of education. Two research questions guide the analysis. First, the report investigates whether male and female students differ significantly in their Mathematics and Language scores. Second, it examines whether scores differ across parental education levels and, if differences are found, which groups differ from each other.

To answer these questions, the analysis uses descriptive statistics, statistical graphics, Welch two-sample t-tests, one-way analysis of variance, Tukey HSD post-hoc tests, and effect size measures. The descriptive analysis provides an overview of the score distributions and group means. The hypothesis tests are then used to assess whether the observed group differences are statistically significant.

The results show statistically significant gender differences in both subjects. Male students had a higher mean Mathematics score, while female students had a higher mean Language score. The results also show significant differences across parental education levels for both subjects. Students in the high school parental education group generally had lower mean scores than students in the higher parental education groups. However, the effect sizes show that parental education explains only a limited proportion of the total variation in scores. Therefore, the findings should be interpreted as statistical associations, not as causal effects.

The rest of the report is organized as follows. Section 2 describes the dataset, variables, data preparation, and research questions. Section 3 explains the statistical methods used in the analysis. Section 4 presents the descriptive results. Sections 5 and 6 report the hypothesis test

results for gender and parental education. Section 7 summarizes the findings and discusses the main limitations of the analysis.

2. Data and Research Questions

The dataset used in this report contains information on student performance in two subjects, Mathematics and Language. According to the task description, both scores are measured on a scale from 0 to 100. The dataset also includes gender and parental level of education. Parental level of education is recorded as an ordinal categorical variable with four levels: high school, associate's degree, bachelor's degree, and master's degree.

The original dataset was stored in long format. This means that each student appeared in two separate rows, one row for the mathematics score and one row for the Language score. The dataset initially contained 972 rows. A check of the student identifiers showed that each of the 486 students appeared exactly twice and had exactly two subject records. Therefore, the data were reshaped into wide format, so that each student formed one observation with separate columns for Mathematics and Language scores.

After reshaping, the final analysis dataset contained 486 observations and five variables: student identifier, gender, parental level of education, Mathematics score, and Language score. No missing values were found in the variables used for the analysis. The score variables were numerical, and all observed scores were within the valid range from 0 to 100. The gender variable contained 245 female students and 241 male students.

The analysis is guided by two research questions. The first research question asks whether male and female students differ significantly in their Mathematics and Language scores. The second research question asks whether scores differ across parental education levels and, if differences are present, which parental education groups differ from each other.

For the first research question, Mathematics and Language scores are each compared between female and male students. For the second research question, Mathematics and Language scores are compared across the four parental education groups. This structure allows the analysis to

answer both the general question of whether differences exist and the more specific question of where those differences occur.

3. Statistical Methods

The statistical methods were selected according to the research questions and the scale level of the variables. Mathematics and Language scores are numerical outcome variables. Gender has two independent groups, while parental level of education has four ordered categories. Therefore, the analysis used descriptive statistics, statistical graphics, Welch two-sample t-tests, one-way ANOVA, Tukey HSD post-hoc tests, and effect size measures. All tests used a significance level of $\alpha = 0.05$.

3.1. Descriptive Statistics and Statistical Graphics

Descriptive statistics were used to summarize the distribution of Mathematics and Language scores before conducting hypothesis tests. For each score variable, the number of observations, mean, median, standard deviation, minimum, and maximum were calculated. The arithmetic mean was used as a measure of central tendency and is defined as (Illowsky & Dean, 2013):

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

where x_i is the score of student i, n is the number of students in the group, and $\bar{x}$ is the average score. In this report, x_i represents either a mathematics score or a language score.

The sample standard deviation was used to measure how strongly scores varied around the mean (Illowsky & Dean, 2013):

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

where s is the sample standard deviation. A larger value of s indicates that the scores are more spread out (Freedman et al., 2007). Descriptive statistics and graphical summaries are commonly used to explore numerical data before formal inference. Mean plots with 95% confidence intervals were used because the research questions focus on differences in average scores between groups, and confidence intervals provide a visual indication of the uncertainty around estimated means.

3.2. Welch Two-Sample t-Test

Welch's two-sample t-test was used to test whether female and male students differed significantly in Mathematics and Language scores. It is appropriate here because gender has two independent groups and the score variables are numerical. Welch's version was chosen because it does not require equal variances between the two groups (Welch, 1947).

The Welch t-statistic is defined as:

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\overline{x_1}$ and $\overline{x_2}$ are the group means, s_1^2 and s_2^2 are the group variances, and $n_1$and n_2 are the group sizes. In this report, group 1 refers to female students and group 2 refers to male students. The null hypothesis states that the female and male mean scores are equal. A p-value below 0.05 was interpreted as evidence of a statistically significant gender difference.

3.3. Cohen's d

Cohen's d was used to measure the size of the gender differences. This is useful because a p-value shows whether a difference is statistically significant, but not how large the difference is. Cohen's d expresses the mean difference in standard deviation units (Cohen, 1988):

$$d = \frac{\overline{x_1} - \overline{x_2}}{s_p}$$

where s_p is the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

In this report, $\overline{x_1} - \overline{x_2}$ represents the difference between the female and male mean scores. Group 1 refers to female students and group 2 refers to male students. Therefore, $n_1 = 245$ is the number of female students and $n_2 = 241$ is the number of male students. The terms s_1^2 and s_2^2 represent the score variances for female and male students, respectively. A negative value of d means that the female mean is lower than the male mean, while a positive value means that the female mean is higher.

3.4. One-Way ANOVA

One-way analysis of variance was used to test whether Mathematics and Language scores differed across the four parental education levels. This method is appropriate because parental education has more than two groups and the outcome variables are numerical. One-way ANOVA tests whether there is a statistically significant difference among several group means by comparing variation between groups with variation within groups (Illowsky & Dean, 2013).

The ANOVA test statistic is:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

In this report, MS_{between} measures the variation between the mean scores of the four parental education groups, while MS_{within} measures the variation among students within the same parental education group. A larger F-value indicates that the group means differ more strongly relative to the variation within groups. The null hypothesis states that all parental education group means are equal. A significant ANOVA result indicates that at least one group mean differs, but it does not identify which specific groups differ. Therefore, Tukey HSD post-hoc tests were used after significant ANOVA results.

3.5. Tukey HSD Post-Hoc Test

When the ANOVA result was significant, Tukey's HSD post-hoc test was used to compare pairs of parental education groups. This test controls the family-wise error rate when several group comparisons are made (Tukey, 1949). The basic Tukey HSD statistic is based on

$$HSD = q_{\alpha,k,N-k}\sqrt{\frac{MS_{\text{within}}}{n}}$$

where $q_{\alpha,k,N-k}$ is the critical value from the studentized range distribution, α is the significance level, k is the number of groups, N is the total sample size, MS_{within} is the within-group mean square from ANOVA, and n is the group size. Since the parental education groups in this dataset had unequal sample sizes, the software implementation used an adjusted version of Tukey's procedure.

3.6. Eta-Squared

Eta-squared was used as an effect size for the ANOVA results. It measures the proportion of total score variation associated with a grouping variable, such as parental education level (Cohen, 1988):

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

where SS_{between} is the variation explained by parental education groups and SS_{total} is the total variation in scores. Larger value of η^2 indicate that the group variable explains a larger proportion of the score variation. Eta-squared was calculated separately for Mathematics and Language.

3.7. Assumption Checks

The t-tests and ANOVA assume independent observations, numerical outcome variables, approximate normality, and acceptable variance homogeneity. Independence was addressed by reshaping the dataset so that each student appeared once in the final analysis dataset. The

outcome variables were numerical scores from 0 to 100. Histograms were inspected to assess the distributions of Mathematics and Language scores. Levene's test was used to assess equality of variances across gender and parental education groups. These checks were used to evaluate whether the planned parametric tests were appropriate.

All analyses were conducted in Python (Python Software Foundation, 2026). The dataset was prepared and reshaped using pandas (McKinney, 2010). Welch t-tests and Levene's tests were conducted using SciPy (Virtanen et al., 2020), while one-way ANOVA and Tukey HSD post-hoc tests were conducted using statsmodels (Seabold & Perktold, 2010). The statistical graphics were created using matplotlib and seaborn (Hunter, 2007).

4. Descriptive Results

Table 1 presents the overall descriptive statistics for Mathematics and Language scores. Both subjects had complete data for all 486 students. The mean Mathematics score was 66.38, while the mean Language score was 69.00. The median was 66.00 for Mathematics and 70.00 for Language. These values show that Language scores were slightly higher than Mathematics scores in the full sample.

Measure	**Mathematics**	**Language**
Count	486	486
Mean	66.38	69.00
Median	66.00	70.00
Standard deviation	15.05	14.45
Minimum	23.00	19.50
Maximum	100.00	100.00

Table 1: Overall descriptive statistics for Mathematics and Language scores

The standard deviation was 15.05 for Mathematics and 14.45 for Language. This means that Mathematics scores were slightly more spread out around the mean than Language scores. Both subjects also covered a wide range of values. The minimum score was 23.00 in Mathematics and 19.50 in Language, while the maximum score was 100.00 in both subjects. All observed scores were within the valid range from 0 to 100.

Figure 1 below shows the mean scores by gender and subject, including 95% confidence intervals. For Mathematics, male students had a higher mean score than female students. Female students had a mean Mathematics score of 63.86, compared with 68.95 for male students. For Language, the pattern was reversed. Female students had a mean Language score of 73.00, compared with 64.93 for male students.

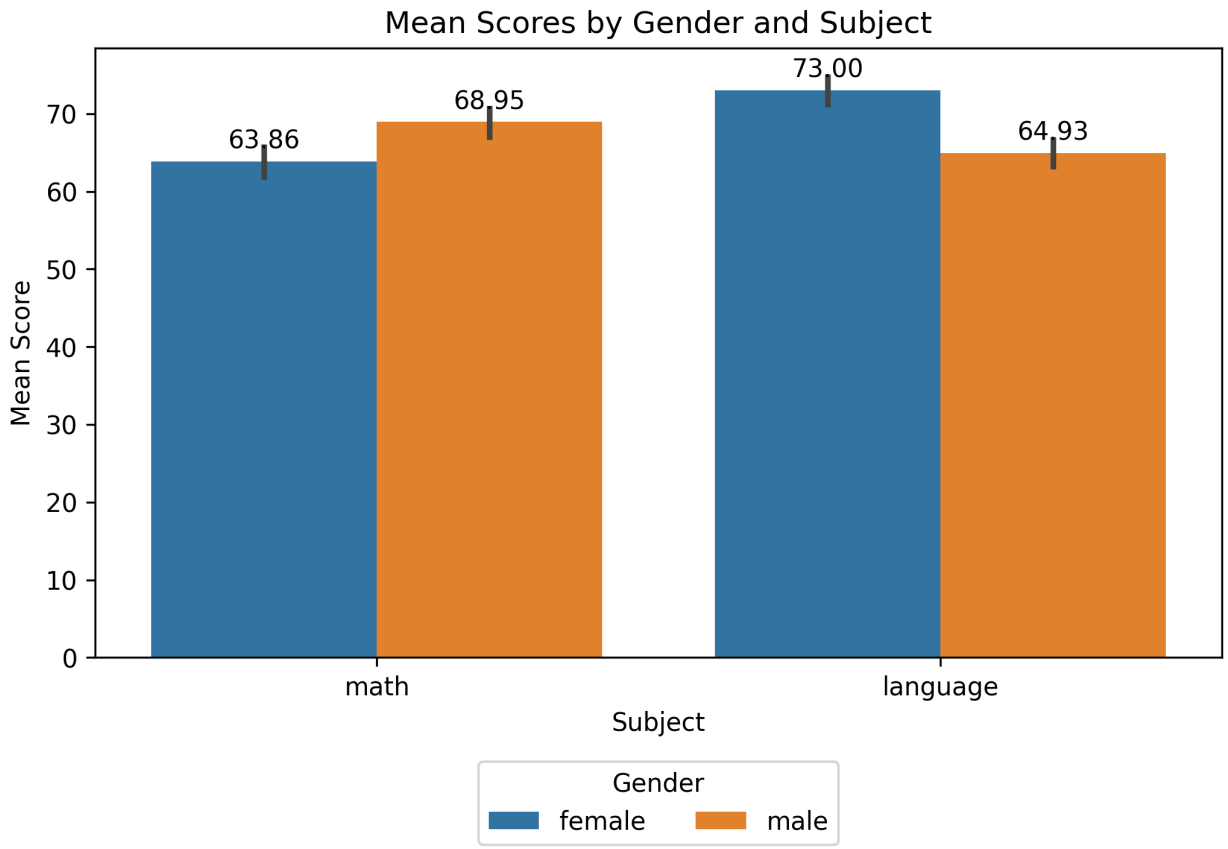


Figure 1: Mean scores by gender and subject

The descriptive results therefore suggest different gender patterns across the two subjects. Male students had higher average scores in Mathematics, while female students had higher average scores in Language. However, these are descriptive results only. Formal hypothesis tests are used in the next section to determine whether the observed differences are statistically significant.

5. Gender Differences in Mathematics and Language

Welch two-sample t-tests were used to examine whether female and male students differed significantly in Mathematics and Language scores. The results are summarized in Table 2 below. The tests were conducted separately for the two subjects.

Research Question	Test used	Statistic	p-value	Effect size	Conclusion
Gender difference in Mathematics	Welch t-test	$t = -3.78$	$p < .001$	$d = -0.343$	Significant
Gender difference in Language	Welch t-test	$t = 6.41$	$p < .001$	$d = 0.581$	Significant
Parental education difference in Mathematics	One-way ANOVA	$F = 6.22$	$p < .001$	$\eta^2 = 0.037$	Significant
Parental education difference in Language	One-way ANOVA	$F = 14.24$	$p < .001$	$\eta^2 = 0.081$	Significant

Table 2: Summary of hypothesis test results

For Mathematics, the Welch t-test showed a statistically significant difference between female and male students, $t(483.23) = -3.78, p < .001$. The negative test statistic reflects the order of comparison, since the female mean score was compared with the male mean score. Female students had a lower mean Mathematics score of 63.86, while male students had a mean score of 68.95. The effect size was $d = -0.343$, which indicates a small to moderate difference.

For Language, the Welch t-test also showed a statistically significant gender difference, $t(484.00) = 6.41, p < .001$. In this case, the positive test statistic indicates that the female mean score was higher than the male mean score. Female students had a mean Language score of 73.00, while male students had a mean score of 64.93. The effect size was $d = 0.581$, which indicates a moderate difference.

These results answer the first research question. There were statistically significant gender differences in both subjects. Male students scored significantly higher in Mathematics, while

female students scored significantly higher in Language. The effect size was larger for Language than for Mathematics, suggesting that the gender difference was more pronounced in Language scores.

6. Parental Education Differences in Mathematics and Language.

One-way ANOVA was used to examine whether Mathematics and Language scores differed across parental education levels. Figure 2 shows the mean scores by parental education level and subject. In both subjects, the mean scores were lowest in the high school parental education group and highest in the master's degree group.

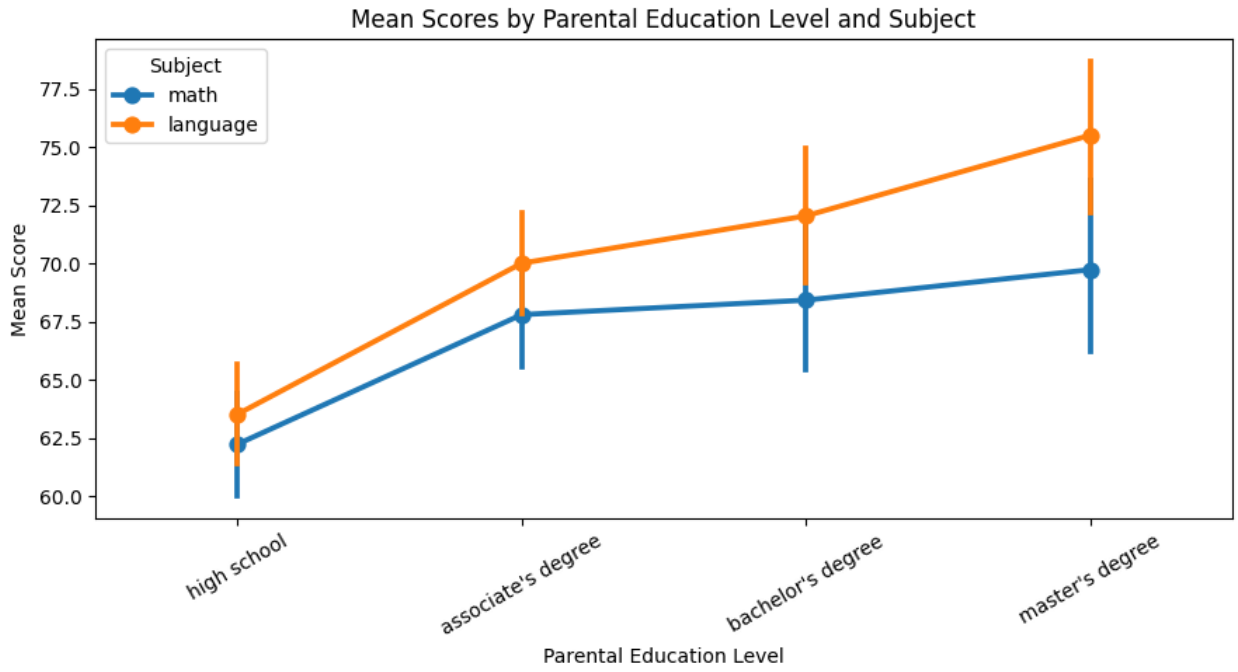


Figure 2: Mean scores by parental education level and subject with 95% confidence intervals

For Mathematics, the ANOVA showed a statistically significant difference across parental education levels, $F(3,482) = 6.22$, $p < .001$. The eta-squared value was $\eta^2 = 0.037$, meaning that parental education explained about 3.7% of the total variation in Mathematics scores. This

indicates a statistically significant but small association between parental education level and Mathematics performance.

For Language, the ANOVA also showed a statistically significant difference across parental education levels, $F(3,482) = 14.24, p < .001$. The eta-squared value was $\eta^2 = 0.081$, meaning that parental education explained about 8.1% of the total variation in Language scores. This suggests that parental education was more strongly associated with Language scores than with Mathematics scores.

Since both ANOVA results were statistically significant, Tukey HSD post-hoc tests were conducted to identify which parental education groups differed from each other. The significant post-hoc comparisons are shown in Table 3 below.

Subject	**Group 1**	**Group 2**	**Mean difference**	**Adjusted p-value**	**CI lower**	**CI upper**
Mathematics	Associate's degree	High school	-5.5700	0.0037	-9.7649	-1.3751
Mathematics	Bachelor's degree	High school	-6.1926	0.0079	-11.1802	-1.2050
Mathematics	High school	Master's degree	7.5022	0.0055	1.6639	13.3404
Language	Associate's degree	High school	-6.4905	0.0001	-10.4226	-2.5585
Language	Associate's degree	Master's degree	5.5028	0.0429	0.1202	10.8854
Language	Bachelor's degree	High school	-8.5265	0.0000	-13.2016	-3.8514
Language	High school	Master's degree	11.9934	0.0000	6.5210	17.4658

Table 3: Significant Tukey HSD post-hoc comparisons

For Mathematics, the Tukey HSD results showed that the high school parental education group differed significantly from the associate's degree, bachelor's degree, and master's degree groups. In each case, the high school group had a lower mean Mathematics score than the higher parental education groups. No significant differences were found among the associate's, bachelor's, and master's degree groups.

For Language, the high school group also differed significantly from the associate's degree, bachelor's degree, and master's degree groups. Again, the high school group had lower mean scores. In addition, the master's degree group had a significantly higher mean Language score than the associate's degree group. No significant difference was found between the associate and bachelor's degree groups or between the bachelor's and master's degree groups.

These results answer the second research question. Mathematics and Language scores differed significantly across parental education levels. The differences were mainly driven by the lower scores in the high school parental education group. However, the eta-squared values show that parental education explained only a limited proportion of the total score variation, especially for Mathematics.

7. Summary and Discussion

This report analyzed whether student performance in Mathematics and Language differed by gender and parental level of education. The analysis was based on a dataset of 486 students. Descriptive statistics, statistical graphics, Welch two-sample t-tests, one-way ANOVA, Tukey HSD post-hoc tests, and effect size measures were used to answer the research questions.

The first research question asked whether male and female students differed significantly in their Mathematics and Language scores. The results showed statistically significant gender differences in both subjects. Male students had a higher mean Mathematics score than female students, while female students had a higher mean Language score than male students. The effect size for Mathematics was small to moderate, while the effect size for Language was moderate. This suggests that the gender difference was more pronounced in Language than in Mathematics.

The second research question asked whether scores differed across parental education levels and, if so, which groups differed from each other. The ANOVA results showed statistically significant differences across parental education levels for both Mathematics and Language. The Tukey HSD post-hoc tests showed that the high school parental education group generally had significantly lower scores than the higher parental education groups. For Language, there was also a significant difference between the associate degree and master's degree groups.

Although the results were statistically significant, the effect sizes should be considered carefully. Parental education explained about 3.7% of the variation in Mathematics scores and about 8.1% of the variation in Language scores. This means that parental education was associated with student performance, but it explained only a limited part of the total variation in scores. Other factors not included in the dataset may also influence student performance, such as study habits, school quality, prior achievement, socioeconomic conditions, teacher support, and language background.

The findings should not be interpreted as causal effects. The analysis shows statistical associations between the available variables and student scores, but it does not prove that gender or parental education directly causes the observed differences. The dataset does not contain information about many other possible explanatory factors. Further analysis could include additional variables and more detailed models to better understand the sources of variation in student performance.

Overall, the analysis found statistically significant differences in both research areas. Gender differences were present in Mathematics and Language, with opposite directions across the two subjects. Parental education differences were also present, mainly because students in the high school parental education group had lower average scores than students in higher parental education groups. The results therefore answer both research questions while also showing that statistical significance should be interpreted together with effect size and study limitations.

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